



Unit 3 · Outcome 2 · Sustainable Development

The case study — aim & strategies

KK01

VCE Environmental Science · Units 3 & 4 · Exam study summary

Key knowledge. · the exact wording “The aim and strategies proposed for addressing the environmental science challenges associ

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



THE IDEA

01 What “a case study” actually means here

Sustainability is a big, abstract idea. A case study is the trick that makes it examinable: you take one real, bounded environmental problem and use it as the testing ground for every concept in the Area of Study.

In VCE Environmental Science, your teacher selects a single case study near the start of this Area of Study. You then return to it for all eight key-knowledge points: its sustainability dimensions, the principles, the challenges, a cost–benefit analysis, the stakeholder tensions, the impacts on Earth’s four systems, and finally how effective the management was. KK01 — this module — is simply the setup: getting the challenge, aim and strategies straight before you analyse anything.

◆ ANALOGY

Analogy · the flight simulator Reading about sustainability is like reading a manual on how to fly. A case study is the flight simulator — one cockpit where every abstract idea (equity, efficiency, trade-offs) suddenly has consequences. You can’t “sort of” manage water: send too much to the wetland and the town runs dry; send too little and the river red gums die. The case study forces the abstract to become concrete.

• NOTE

Watch the wording — two different “aims” The aim of the case study (the goal of the real-world management response) is not the same as the aim of your scientific investigation (your poster in Area of Study 3). This dot point is about the management response. Mixing them up is a classic mark-loser.

THE SELECTED CASE STUDY

02 The Boorla Wetlands recovery, Yarrin River

A teaching case study set in regional Victoria. It’s fictional, but every part of it mirrors real Murray–Darling-style water management you could meet in the exam.

The Yarrin River is a regulated lowland river. Yarrin Dam upstream stores water for irrigation and town supply, so flow downstream is controlled by people, not just rain.

The Boorla Wetlands (≈4,000 ha) are a Ramsar-listed floodplain: river red gum forest, migratory waterbirds, Murray cod and golden perch. They need periodic flooding to stay healthy.

Connawarra (pop. ≈6,000) and a district of irrigated dairy and horticulture depend on the same river for jobs, food production and drinking water.

The Boorla people hold cultural connection to the wetland and its water, including culturally significant sites that depend on flooding.

Notice the design of the case: one finite resource (water) that four groups all need, for different reasons. That overlap is what makes it a genuine sustainability problem rather than a simple engineering job.

THE CHALLENGE

03 The environmental science challenge

The “challenge” is the problem — what is going wrong, and what makes it hard to fix. It is not the goal and it is not the actions. Pin it down precisely.

- **NOTE**

The Boorla challenge — in one breath Decades of upstream water extraction, combined with a hotter, drier climate, have cut the volume and altered the timing of flows reaching the Boorla Wetlands — driving river red gum dieback, collapsed waterbird breeding, falling native fish numbers and rising salinity — while the very same water underpins the jobs, food production and town supply of Connawarra and the cultural water rights of the Boorla people.

Two halves make it a challenge and not just a problem: an ecological decline (the wetland is dying) and a human dependence on the same resource (you can't just "add water" without taking it from someone). Manage one and you threaten the other. That tension is the whole reason this Area of Study exists.

- ◆ **ANALOGY**

Analogy · the GP and the patient Think of a doctor with a patient who has high blood pressure but works night shifts and can't easily change their routine. The challenge is the illness plus the patient's constraints. The aim is the treatment goal ("get blood pressure into a safe range without a plan they can't stick to"). The strategies are the specific interventions (medication, diet, exercise, regular check-ups). Nobody confuses the goal with the tablets — and examiners expect the same clarity from you.

THE AIM

04 The aim of the management response

The aim is the destination — a single, clear statement of what success would look like. A strong sustainability aim almost always has two halves: restore the environment while protecting the people who depend on it.

- **NOTE**

The Boorla aim "To restore the ecological health of the Boorla Wetlands — re-establishing river red gum condition, waterbird breeding and native fish populations — while maintaining a secure and fair water supply for the Connawarra community, irrigators and Traditional Owners, both now and into the future."

If the aim were only "restore the wetland", you could achieve it by simply cutting off every farm — but that destroys the community. If it were only "keep water for the town and farms", the wetland dies. The word while bakes the trade-off into the goal. The hint of "now and into the future" and "fair" quietly foreshadows the sustainability principles you'll meet in KK03 (intergenerational and intragenerational equity) — but here you just need to recognise a well-formed aim.

Specific actions to reach the aim. Usually several.

THE STRATEGIES

05 The strategies proposed

Strategies are the actions — the deliberate interventions chosen to move toward the aim. A good answer names the strategy and then says how it tackles the challenge. A name on its own is half an answer.

Targeted releases from Yarrin Dam timed to mimic natural spring floods, triggering river red gum growth and waterbird breeding.

Lining earthen channels, drip irrigation and metering so farms grow the same crops with less water — the savings returned to the river.

Government purchases water entitlements from willing irrigators, adding that volume to the environmental allocation.

Treating and reusing Connawarra's wastewater for parks and some irrigation, cutting fresh extraction from the river.

Revegetating the floodplain and pumping shallow groundwater to lower the water table and slow salt build-up.

Water allocated to and co-managed with the Boorla people to restore culturally significant sites.

• **NOTE**

The pattern examiners reward Name + mechanism + link. "Environmental flows (name) deliver timed dam releases that mimic natural floods (mechanism), which re-triggers red gum growth and bird breeding that had stopped because the wetland no longer floods (link to the challenge)." That single sentence is worth full marks where a bare list isn't.

INTERACTIVE · 3D

06 Flow simulator — feel the trade-off

This is the "environmental flows" strategy made tangible. Slide the dial to send more environmental water to the wetland and watch it recover — but watch what happens to the water left for the town and farms. There is no free lunch; that's the point.

• **NOTE**

What this scene is teaching One strategy can't be judged in isolation. Environmental flows clearly help the ecological side, but every gigalitre sent to the wetland is a gigalitre the economic and social sides don't get. That's exactly why the case uses several strategies together — efficiency and recycling free up water so the wetland can be fed without simply taking it from the town.

INTERACTIVE · SKILL DRILL

07 Case-study anatomy — challenge, aim or strategy?

The single most common exam slip is mislabelling these three. Read each statement and tag it. You'll get instant feedback and a running score.

EXAM TECHNIQUE

08 Command words — what the verb is asking

◆ **COMMAND WORDS**

For this dot point you'll mostly meet six command words. The verb tells you how much to write and in what shape. Tap to reveal.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

WRITING

09 P·E·L trainer — build a model answer



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

P·E·L WRITING

P — Point: name your strategy / claim. E — Explain: how it works / the mechanism. L — Link: tie it back to the challenge or aim in the case study. Most full-mark sentences are one clean P·E·L.

PITFALLS

10 Exam traps for this dot point

EXAM TRAPS

Describing what's going wrong when asked for the aim. The challenge is the problem; the aim is the goal. "The wetland got healthier" is a result, not a strategy. A strategy is an action you choose to take. Naming a strategy with no mechanism or link earns half marks at best. Giving only the ecological half and dropping the "while keep water secure" half loses the balance the aim needs. Writing about "the environment" in general instead of this case study. Examiners reward specifics. Confusing the case-study aim with your scientific-investigation aim from Area of Study 3.

EXAM PRACTICE

11 Practice questions — weak vs strong

Cover the answers, attempt each question yourself, then reveal. Each one shows a realistic weak answer, an annotated strong answer, a marking guide, and a self-mark. Your score collects in the dock at the bottom-right.

Identify the environmental science challenge in the Boorla case study, and state the aim of the management response.

- 1 — challenge identified as ecological decline of the wetland plus human dependence on the same water.
- 1 — aim stated as restoring wetland health while maintaining secure/fair water supply.

Outline three strategies proposed to address the Boorla challenge. For one strategy, explain how it helps address the challenge.

- 1 — three legitimate strategies named (e.g. environmental flows, efficiency upgrades, buy-backs, recycled water, salinity management, cultural flows).
- 2 — for one strategy: the mechanism (how it works) and a link to the challenge it addresses.

Distinguish between the aim and a strategy in an environmental case study. Use examples from the Boorla case to support your answer.

- 1 — aim correctly characterised as the overall goal / desired outcome.
- 1 — strategy correctly characterised as a specific action to achieve the aim.
- 1 — a genuine point of contrast (goal vs action, or one aim vs several strategies).
- 1 — supported with at least one correct Boorla example of each.

Read the stimulus, then summarise the aim and describe two strategies it proposes.

• **NOTE**

Stimulus · adapted management brief “The Marra Lakes program seeks to bring breeding pelican colonies back to the lakes while protecting the fruit-growing district that draws on the same channel. Managers plan to deliver a managed flood every two years using stored water, and to convert 1,200 ha of open irrigation channel to piped delivery to cut evaporation losses.”

- 2 — aim summarised with both halves (restore pelican breeding and protect the district).
- 1.5 — strategy 1 described (managed biennial flood) and what it achieves.
- 1.5 — strategy 2 described (piping channels) and what it achieves.

Explain why a single, well-defined case study is used to study sustainable development, and describe how the aim and strategies of the Boorla case study connect to its environmental science challenge.

- 2 — why a single case study: makes the abstract concrete/bounded; allows consistent, comparable analysis across the Area of Study.
- 1 — the Boorla challenge accurately restated (two-sided).
- 1.5 — clear link: the aim mirrors the two sides of the challenge.
- 1.5 — strategies mapped to the aim/challenge (some repair ecology, others free up water for people).

U3.02 · KK01 SUSTAINABLE DEVELOPMENT

The case study — aim & strategies

Every other dot point in this Area of Study leans on one chosen case study. This module sets it up: the real-world challenge being managed, the aim of the response, and the strategies proposed to get there — and, crucially, how to tell those three apart in an exam.

“The aim and strategies proposed for addressing the environmental science challenges associated with the selected case study.”

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

The challenge is the environment is being damaged. The aim is to fix the wetland.” Vague on both. “The environment is damaged” doesn’t name the conflict between ecology and water use, and “fix the wetland” drops the human half of the aim. ≈0–1 mark.

✓ STRONG / MODEL ANSWER

The challenge is that reduced river flows (from extraction and a drying climate) are degrading the Boorla Wetlands, yet the same water sustains Connawarra’s farms, town supply and the Boorla people’s cultural water. The aim is to restore the wetland’s ecological health while keeping water supply secure and fair, now and into the future.” Names the two-sided conflict (1) and a two-sided aim (1). 2/2.

✗ WEAK ANSWER

Use less water, plant trees, and recycle. Planting trees is good for the environment.” Three loosely-named ideas, but the explanation is generic (“good for the environment”) with no mechanism and no link to the challenge. ≈1 mark.

✓ STRONG / MODEL ANSWER

Three strategies: (i) environmental flows, (ii) water-use efficiency upgrades and (iii) water buy-backs. Environmental flows deliver timed dam releases that mimic natural spring floods; this re-triggers river red gum growth and waterbird breeding that had stopped because the wetland no longer floods — directly reversing the ecological decline at the heart of the challenge.” Three valid named strategies (1) + a clear name-mechanism-link explanation of one (2). 3/3.

✗ WEAK ANSWER

The aim and the strategy are both about fixing the wetland. The aim is to fix it and the strategy is also to fix it.” No real distinction drawn, and the examples just repeat “fix it”. “Distinguish” needs a clear contrast on a shared feature. ≈1 mark.

✓ STRONG / MODEL ANSWER

An aim is the overall goal — the single statement of what success looks like — whereas a strategy is one specific action chosen to move toward that aim. At Boorla, the aim is to restore wetland health while keeping water secure and fair; a strategy such as environmental flows is just one of several actions used to achieve it. There is one aim but several strategies, and a strategy can succeed or fail without changing the aim itself.” Defines each (2), contrasts goal-vs-action and one-vs-many (1), and grounds both in Boorla examples (1). 4/4.